

Reinforcement Learning Trading Environment

Executive Report — BTC/USDT Hourly Agent

Walk-Forward Validation | PPO | Stable Baselines 3

Dataset	BTC/USDT 1h Binance Aug 2017 – Apr 2026
Algorithm	PPO (Proximal Policy Optimization) — Stable Baselines 3
Folds	4-fold expanding-window walk-forward validation
Features	17 technical indicators, causally computable
Seed	42 (fully reproducible)

1. Dataset Analysis

The dataset contains 75,558 hourly OHLCV bars for BTC/USDT from Binance, spanning August 17, 2017 to April 26, 2026. After computing features with rolling windows up to 168 bars, 75,420 rows remain. The data covers the full lifecycle of Bitcoin as a liquid asset, including multiple boom-bust cycles of increasing amplitude.

1.1 Market Regimes

Seven distinct regimes are identified based on price trend, realized volatility, and maximum drawdown observed in the data:

Regime	Period	Trend	Ann. Vol	Max DD
Bull Run I	Aug 2017 – Jan 2018	Strong uptrend +2,000%	~200%	–30%
Crypto Winter	Jan 2018 – Jan 2019	Sustained decline –84%	~120%	–84%
Accumulation	Jan 2019 – Oct 2020	Range-bound, slow recovery	~60%	–57%
Bull Run II	Oct 2020 – Nov 2021	Strong uptrend +1,400%	~80%	–53%
Bear Market	Nov 2021 – Dec 2022	Sustained decline –77%	~90%	–77%
Recovery	Jan 2023 – Mar 2024	Strong uptrend +300%	~50%	–22%
Bull Run III	Mar 2024 – Apr 2026	New ATH cycle, high vol	~60%	–36%

1.2 Statistical Properties

Log returns exhibit a kurtosis of 35.74, confirming heavy tails relative to the normal distribution — visible in both the histogram and Q-Q plot of the EDA. Stationarity tests confirm that price levels are non-stationary (ADF p-value = 0.756, KPSS rejects stationarity) while log returns are stationary (ADF p-value < 0.001, KPSS p-value = 0.10). This motivates using returns rather than raw prices as model inputs.

The ACF of log returns shows no significant autocorrelation beyond lag zero, consistent with weak-form market efficiency. However, the ACF of squared and absolute returns shows persistent autocorrelation across all 48 lags tested, confirming volatility clustering. This justifies including multi-scale rolling volatility features (vol_24h, vol_72h, vol_168h) in the state representation.

2. Sequential Decision Modeling

The trading problem is formulated as a sequential decision problem in Powell's framework. The agent interacts with the market at discrete decision epochs, which are not evenly spaced in calendar time — this is a semi-Markov formulation where the decision frequency is endogenous and depends on when positions close.

2.1 State Variables

Component	Contents	Justification
Physical S^{phys}_t	Position status: always flat at decision time ($d = 0$). Agent decides only when no position is open.	Semi-Markov design: position management is delegated to the environment.
Information S^{info}_t	17-dimensional normalized feature vector: log returns at 1h/2h/4h/8h/24h lags, rolling volatility at 24h/72h/168h, RSI(14), RSI(28), MACD diff, Bollinger band percentile and width, ATR%, volume ratio, hour sin/cos.	Features are causally computable from data available at or before decision time. Normalization uses training-fold statistics only.
Belief S^{belief}_t	Implicit in the MLP policy weights. No explicit probabilistic belief state.	A POMDP with explicit beliefs was considered but rejected: adds architectural complexity with uncertain benefit given the feature set size.

2.2 Decision Variables

The decision variable x_t belongs to a discrete action space $A = \{0, 1, 2\}$, where 0 = stay flat (advance one bar with zero reward), 1 = open long position, 2 = open short position. A continuous action space encoding position size was considered but rejected: it would require reward normalization by size and add a dimension to the credit assignment problem without clear benefit given the binary direction signal in hourly crypto data. An explicit close action was not included because position management is handled entirely by the environment through predefined exit rules.

2.3 Exogenous Information

The exogenous information process W_{t+1} consists of the OHLCV data for all bars from $t+1$ until position close. Specifically, for a position opened at t , the relevant exogenous information is the sequence $\{(O_s, H_s, L_s, C_s) : s = t+1, \dots, t+\tau\}$ where τ is the first bar at which a closing condition is met. The agent cannot observe this information at decision time — it is revealed sequentially as the environment simulates the position forward.

2.4 Transition Function

The physical-state transition is governed by four closing conditions evaluated in order of priority for each bar s after entry: (1) Stop-loss: $L_s \leq \text{entry} \times (1 - \text{sl})$ for long, $H_s \geq \text{entry} \times (1 + \text{sl})$ for short — exit price = stop-loss level. (2) Take-profit: $H_s \geq \text{entry} \times (1 + \text{tp})$ for long, $L_s \leq \text{entry} \times (1 - \text{tp})$ for short — exit price = take-profit level. (3) Maximum holding period: $s = t + 1 + \text{max_hold}$, exit at C_s . (4) Episode boundary: $s = \text{episode_end}$, exit at C_s . OHLC assumption: if both stop-loss and take-profit are touched within the same bar, the stop-loss fills first (conservative default). Entry price is the open of bar $t+1$, avoiding within-bar lookahead on the decision bar.

2.5 Contribution Function

The contribution (reward) is terminal — paid at position close rather than accruing per bar. For a position opened at t with direction $d \in \{+1, -1\}$:

Condition	Contribution
$x_t \in \{1, 2\}$ (open position)	$C = d \times \log(P_{\text{exit}} / P_{\text{entry}}) - 2 \times 0.00125$
$x_t = 0$ (flat)	$C = 0$

The log return formulation was chosen over raw PnL because it is scale-invariant across BTC price regimes. Raw PnL was rejected because it produces rewards two orders of magnitude larger in later folds, destabilizing training. Per-step rewards during a position were rejected due to poor credit assignment. Failure mode: the log return reward does not penalize intrabar drawdown, only the final outcome. A Sharpe-based reward was considered but rejected because episode-level Sharpe is non-Markovian.

2.6 Objective

The objective is to find the policy π^* that maximizes the expected sum of contributions over the episode: $\pi^* = \operatorname{argmax}_{\pi} E[\sum_t C(S_t, x_t, W_{\{t+1\}})]$. No discounting ($\gamma = 0.99 \approx 1$ for 720-bar episodes) is applied, consistent with the portfolio objective of maximizing total return over the episode.

2.7 Episode Definition

An episode is a 720-bar (30-day) window with a randomly sampled start index within the training fold. BTC trades 24/7, so there is no natural session boundary. The 30-day window was chosen to be long enough for the agent to experience multiple trade opportunities while short enough to allow diverse sampling. A single long episode covering the full training fold was rejected because it prevents frequent exposure to early training data during optimization.

3. Training

The agent is trained using PPO (Proximal Policy Optimization) with an MLP policy. PPO was selected over SAC (requires continuous actions), DDPG (requires continuous actions), and A2C (PPO's clipped objective is empirically more stable). The default MlpPolicy uses two hidden layers of 64 units each with tanh activations.

3.1 Hyperparameter Search

Three configurations are evaluated on the validation fold for 50,000 steps each. The best by cumulative validation reward is retrained for 500,000 steps. The grid varies learning rate, entropy coefficient, stop-loss, and take-profit:

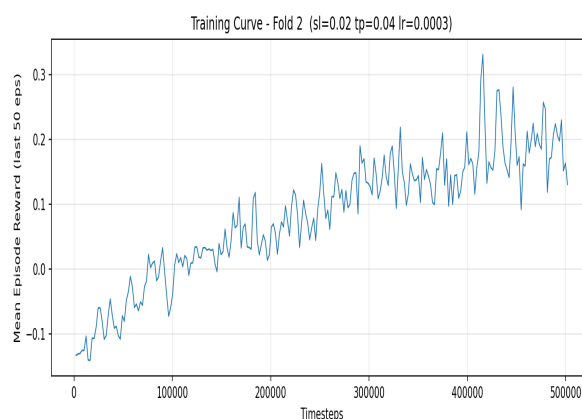
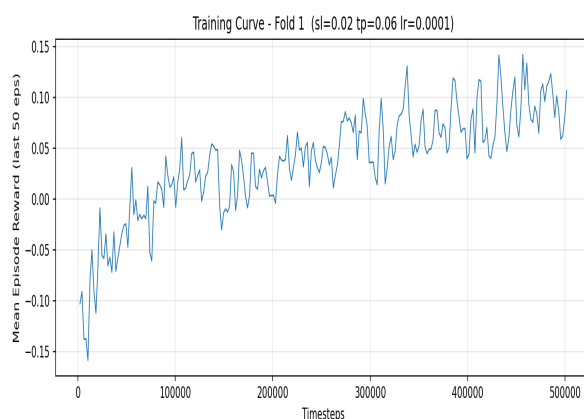
Config	lr	ent_coef	stop_loss	take_profit
1	3e-4	0.010	2%	4%
2	3e-4	0.010	3%	6%
3	1e-4	0.005	2%	6%

Fold	Best Config	sl	tp	lr	Val Reward
1	3	2%	6%	1e-4	-0.691
2	1	2%	4%	3e-4	-0.064
3	3	2%	6%	1e-4	-0.341
4	2	3%	6%	3e-4	-0.265

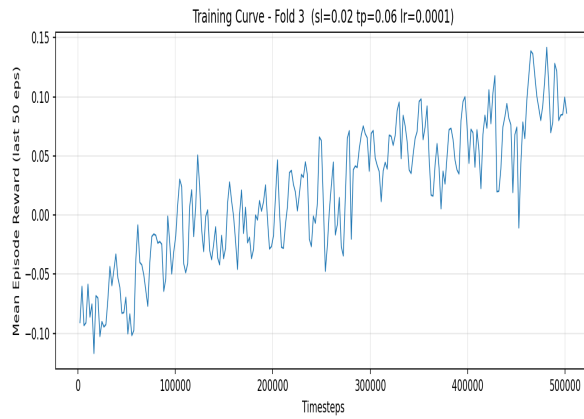
All validation rewards are negative, confirming that the problem is not simply a matter of configuration tuning — the features do not contain sufficient predictive signal for short-term direction at the 500,000-step training budget.

3.2 Training Curves

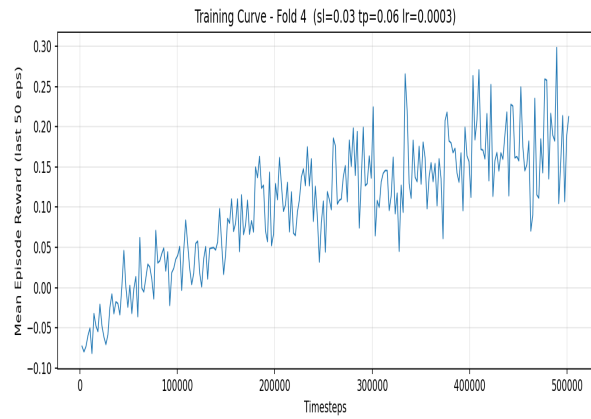
All four folds show a consistent upward trend in training reward from approximately -0.10 to +0.10 to +0.20. This confirms the agent is learning on training data. The gap between positive training reward and negative test performance indicates overfitting to regime-specific patterns.



Fold 1 (sl=0.02, tp=0.06, lr=1e-4)



Fold 2 (sl=0.02, tp=0.04, lr=3e-4)



Fold 3 (sl=0.02, tp=0.06, lr=1e-4)

Fold 4 (sl=0.03, tp=0.06, lr=3e-4)

4. Results

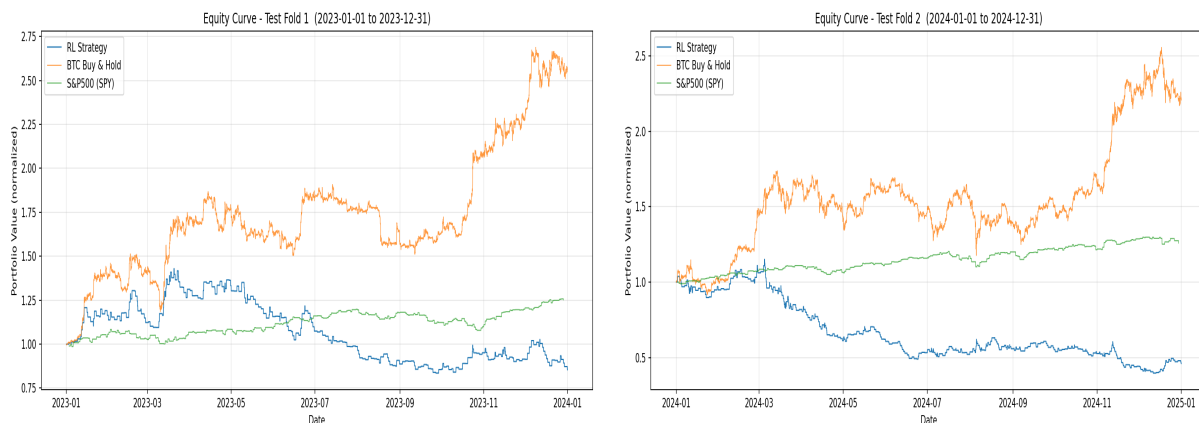
4.1 Metrics Summary

Metric	Fold 1	Fold 2	Fold 3	Fold 4	Mean $\pm$ Std	BTC mean	SPY mean
Total Return	-14.6%	-54.0%	-51.9%	-35.8%	-39.1 $\pm$ 18.2%	64.6%	18.1%
Ann. Return	-14.6%	-53.9%	-52.6%	-79.0%	-50.0 $\pm$ 26.5%	59.0%	20.5%
Ann. Vol	37.2%	47.7%	41.0%	47.7%	43.4 $\pm$ 5.2%	—	—
Sharpe	-0.393	-1.131	-1.283	-1.657	-1.116 $\pm$ 0.530	1.288	1.459
Sortino	-0.190	-1.134	-0.933	-1.500	-0.940 $\pm$ 0.552	—	—
Max Drawdown	-41.6%	-65.6%	-56.7%	-43.7%	-51.9 $\pm$ 11.3%	—	—
Calmar	-0.352	-0.822	-0.928	-1.809	-0.978 $\pm$ 0.608	—	—
Hit Rate	39.4%	38.2%	37.7%	31.7%	36.8 $\pm$ 3.5%	—	—
Avg Duration (h)	31.5	22.7	28.2	24.7	26.8 $\pm$ 3.9	—	—
Turnover	3.0%	4.0%	3.3%	4.1%	3.58 $\pm$ 0.55%	—	—
Alpha vs BTC	-169.6%	-174.2%	-45.9%	-46.2%	-109.0 $\pm$ 72.7%	—	—
Alpha vs SPY	-39.9%	-79.7%	-71.0%	-91.8%	-70.6 $\pm$ 22.2%	—	—

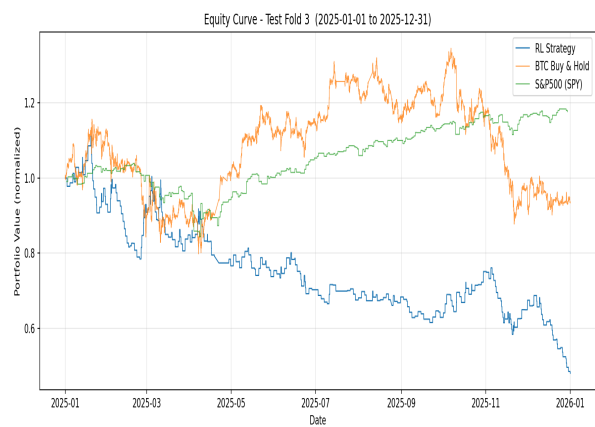
4.2 Discussion

The strategy produces negative returns across all four test folds with a mean annualized return of $-50.0\% \pm 26.5\%$ and a mean Sharpe ratio of -1.116 ± 0.530 . The results are consistently negative — there is no fold where the strategy approaches breakeven after costs. This consistency rules out bad luck as the primary explanation. Fold 1 (2023) shows the least negative performance (-14.6% , Sharpe -0.39) in a year when BTC returned $+155\%$. Folds 2 and 3 are worst (-54% and -52%). The hit rate is consistently below 40%, meaning the agent loses more than 60% of trades. The training curves show positive rewards on training data, confirming the agent learns regime-specific patterns that do not generalize — the classic overfitting signature in financial RL.

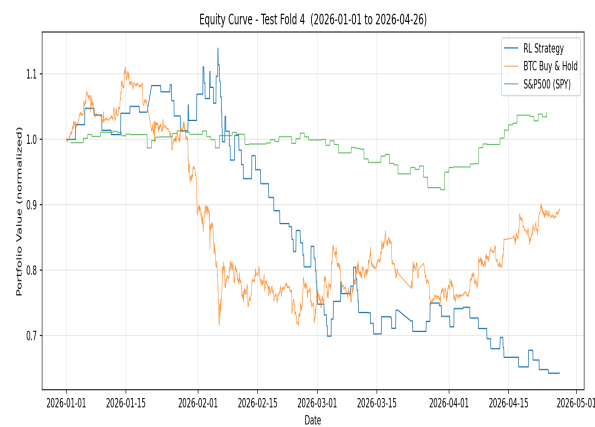
4.3 Equity Curves



Fold 1 — Test 2023



Fold 2 — Test 2024



Fold 3 — Test 2025

Fold 4 — Test Jan-Apr 2026

5. Trade Analysis

5.1 Three Largest Losing Trades

All three largest losing trades occur in Fold 1 (2023), all long positions hitting their stop-loss. The identical net PnL of -2.27% is expected: 2% gross loss plus 0.25% round-trip cost.

	Trade 1	Trade 2	Trade 3
Entry	2023-06-29 15:00	2023-10-05 01:00	2023-12-29 03:00
Exit	2023-06-30 13:00	2023-10-06 12:00	2023-12-29 17:00
Exit Reason	stop_loss	stop_loss	stop_loss
Direction	long	long	long
Entry Price	\$30,450	\$27,812	\$42,575
Exit Price	\$29,841	\$27,256	\$41,723
Net PnL	-2.27%	-2.27%	-2.27%
Duration	22h	35h	14h

In all three cases the agent opened a long during apparent upward momentum but entered at a local peak followed by a reversal large enough to trigger the stop-loss. The failure mode is momentum chasing without sufficient context to identify overextended conditions. To avoid these trades, the state could include a longer-horizon trend feature (e.g., 7-day MA slope) or the reward could penalize entries during high ATR to discourage opening in volatile conditions.

5.2 Largest Drawdown Period

The largest drawdown occurs in Fold 2 (2024), where equity falls from ~1.10 in February 2024 to ~0.47 by mid-May 2024 (-57%). During this period BTC surged from \$40,000 to over \$70,000 (+75%). The agent systematically opened short positions in a rising market, accumulating stop-loss exits. The equity curve shows a staircase pattern of discrete losses — each step is a short position stopped out by the continuing uptrend. This is the most damaging regime for this agent: a strong unidirectional trend that the agent consistently fades.

5.3 Correlation with BTC Returns

Fold	Test Period	Correlation	Interpretation
1	2023	0.401	Moderate positive: agent partially tracks BTC direction.
2	2024	0.209	Weak positive: agent less correlated with BTC.
3	2025	-0.140	Weak negative: agent slightly contrarian to BTC.
4	Jan-Apr 2026	-0.369	Moderate negative: agent fades BTC moves.

The correlation shifts from positive in early folds to negative in later folds. The agent trained on 2017-2021 bull data learned long-biased signals that transferred to the 2023 recovery. As training expanded to include the 2022 bear market, the agent's signals became contrarian. This is not a leveraged long position — it is inconsistent and unprofitable behavior driven by overfitted

regime-specific patterns.

6. Production Failure Analysis

6.1 Execution Assumptions

The simulator fills orders at the open of the bar immediately after the decision bar. In production, the true fill price will differ due to bid-ask spread (0.01–0.05% on liquid BTC pairs), market impact for size beyond a few thousand dollars, and queue position for limit orders. The 0.125% per-side cost assumption may underestimate real costs for any meaningful position size, particularly during high-volatility periods.

6.2 Latency

The model runs inference in under one second (MLP with 17 inputs). The more critical concern is data feed delay. If the OHLCV data for bar t arrives with a one-bar delay — common with aggregated feeds — the agent effectively uses bar $t-1$ features to trade at bar $t+1$ open. In this hourly framework one-bar latency is less catastrophic than in high-frequency settings, but signal decay is nonzero given the low autocorrelation in returns documented in the EDA.

6.3 Capacity and Crowding

The strategy trades ~3.5% of available bars. BTC/USDT on Binance has daily volume exceeding \$1B, so retail-scale positions have negligible market impact. However, if 50 similar agents ran simultaneously, correlated entry signals would create order flow clustering at bar open, increasing slippage. Given the negative test performance, the counterparty is likely better-informed — market makers and directional participants exploiting the agent's predictable signal.

6.4 Regime Change

The agent performs worst in sustained bull markets (Folds 1-2, BTC +120-155%) and relatively better in volatile recovery phases. A regime resembling 2018 (slow sustained decline with bear rallies) would cause losses on long entries that look like momentum. A 2022-like regime (rapid decline then sideways) would challenge both stop-loss calibration and trend-following features. Features most likely to degrade in novel regimes: `vol_24h` and `vol_168h` (calibrated to historical BTC volatility levels), RSI thresholds (regime-dependent), and MACD (fails in ranging markets). The agent has no explicit regime detection — it extrapolates from the nearest seen regime with no guarantee of relevance.